

Multi-Degree-of-Freedom Planar Linkage Synthesis for Leg Designs

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Abstract—Linkage synthesis methods help design and tune robot legs for specific applications. This paper proposes a topological and dimensional linkage synthesis tool for multi-degree-of-freedom (DoF) planar linkages with revolute joints. It enumerates a large number of topologies with analytical forward kinematics and synthesizes their dimensions with gradient-based optimization. The optimization not only supports a variety of costs and constraints for specifying design requirements, but is also fast and scalable with parallel computation on graphics processing units. Three leg examples, including a one-DoF leg for walking mechanisms, a two-DoF parallel leg with a fan-shaped workspace, and a three-DoF hybrid leg with an active ankle, are designed and validated in the real world. This tool also has potential for grippers, wings, arms and more.

I. INTRODUCTION

Linkages are widely used in legged robots thanks to their versatility, strength, and relatively simple fabrication. Although many designs have already been employed in legged robots, methods for linkage synthesis remain necessary when designing legs for specific tasks and new scenarios. For example, the designs may need to satisfy specific requirements for foot trajectory, workspace, and Jacobian. Meanwhile, they also need to be compact and lightweight, and their workspaces should not contain singularities. These design objectives become more challenging to reach and balance when multiple degrees of freedom (DoFs) are involved.

Many linkage synthesis methods have already been proposed, covering a wide range of topologies with different DoFs and offering options for determining their dimensions based on various objectives. However, they are scattered across multiple sources, use different mathematical formulations, apply to a relatively small set of topologies, and support different design objectives and constraints. They may also not include the necessary code or software to help others get started quickly.

Motivated by our own research need to expand the design space of robot legs for task-specific designs, this paper proposes a topological and dimensional linkage synthesis tool for multi-DoF planar linkages with revolute joints as shown in Fig. 1. We chose to build a tool for this specific category of linkages because they are easy to fabricate, can already generate a variety of behaviors, and are widely adopted in legged robots. While synthesis methods for spherical and spatial linkages may apply to planar cases as well, we

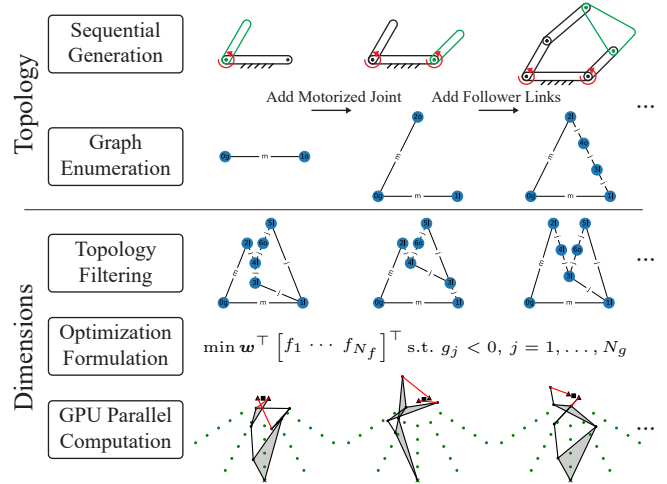


Fig. 1. Pipeline of the proposed Multi-DoF planar linkage synthesis tool for leg designs. The source code is available at <https://github.com/iicfcii/linkage-leg-code>.

decided to develop a specialized method for simplicity and performance.

The unique features of the proposed tool are: 1) support of a large number of multi-DoF series, parallel, and hybrid linkage topologies; 2) compatibility with a wide range of costs and constraints for specifying and tuning designs; 3) fast and scalable optimization on graphics processing units (GPUs); and 4) open-source implementation in Python for ease of use and tweaking.

II. BACKGROUND

Planar linkages are widely used in legged robots. They can be employed to generate prescribed trajectories across multiple feet [1], [2]. They are helpful for remotely actuating joints to locate motors proximally and reduce leg inertia [3], [4]. Their carefully tuned, varying gear ratios between the input and output motions can be leveraged for various functions, such as increasing series elastic energy storage during jumping [5] and flattening the torque requirement at the knee joint [6]. Moreover, multi-DoF parallel designs enable load sharing among actuators [7], [8].

Synthesizing planar linkages has two major steps: topological and dimensional syntheses [9]. Topological synthesis determines the required number of links, the number and types of joints, and the connection structures to achieve desired DoFs. Dimensional synthesis then determines the dimensions or proportions of the links of a linkage.

When only synthesizing topologies, graph theory and combinatorial analysis are usually used to enumerate the topologies of interest [9]. Composition of Assur Graphs is another way to construct topologies [10], [11]. For a fixed

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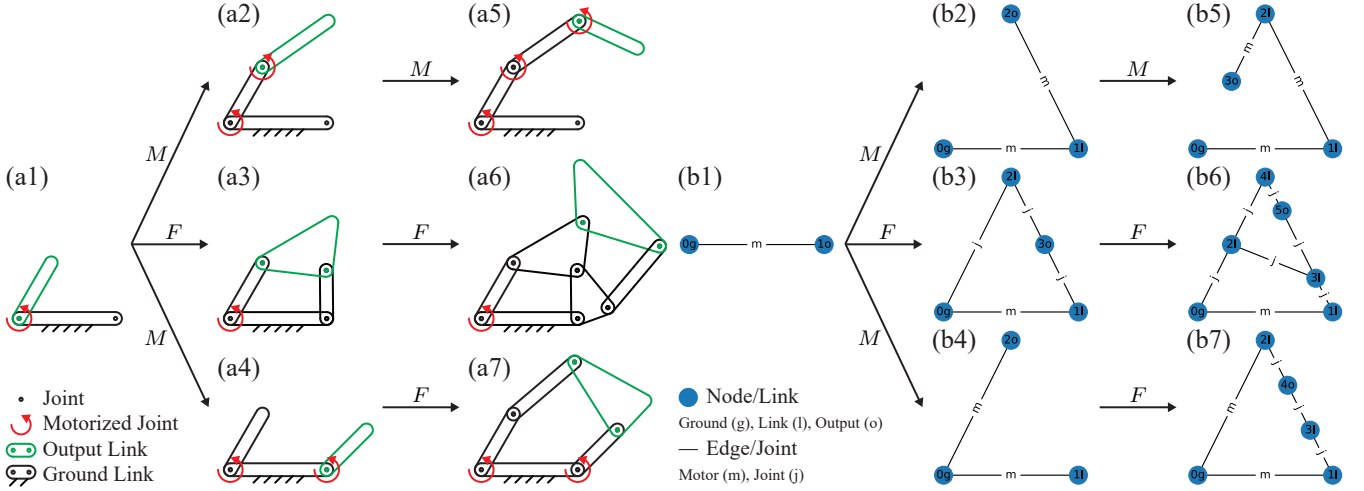


Fig. 2. Topology generation with two operations: adding a motorized link M and adding a pair of follower links F . (a1)-(a7) Example linkage topologies and how they are generated. (b1)-(b7) Graph representations of those linkage topologies.

set of topologies, many dimensional synthesis methods or tools have been developed. For example, MotionGen can synthesize planar four-bar linkages with revolute and prismatic joints, supporting mixed exact and approximate pose constraints as well as line and point constraints for pivots [12]. Synthesis of one-DoF six-bar linkages with revolute joints has also been tackled in a series of studies, though some post-processing is required to identify the defect-free designs [13], [14], [15]. For two-DoF five-bar linkages with revolute joints, closed-form synthesis equations to produce specified velocity ellipses at selected points in their workspace have been derived, but they do not enforce any practicality constraints [16]. It has also been demonstrated that the five-bar synthesis can be formulated as an optimization problem, providing more flexibility in specifying the desired behavior through costs and constraints [17].

Some methods do not assume the set of topologies. For example, LINKS is a dataset of 100 million one-DoF planar linkages with revolute joints and 1.1 billion coupler curves [18]. The topologies are generated through repeatedly adding a pair of links connected via a revolute joint to two existing joints of the previous linkage. The benefit of this approach is that the forward kinematics is analytical, making it fast to solve. Thanks to the large amount of data, linkages for a target path can even be found by a simple searching algorithm without any optimization. Another synthesis approach involves starting with a fully actuated serial chain of links performing a prescribed cyclic motion, and iteratively replacing the joint motors with coupling links connected to nearby links, ultimately yielding a one-DoF linkage that closely reproduces the motion [19]. The major limitation of these methods is that they only support one-DoF designs.

Driven by the idea of determining the topology and dimensions of planar linkages simultaneously instead of treating them as separate steps, methods based on the spring-connected rigid block model paired with gradient-based optimization have also been proposed [20]. Their successful applications to robotic devices, including a transformable wheel [21] and a knee exoskeleton [22], have demonstrated

powerful capabilities for synthesizing linkages with both revolute and prismatic joints and multiple passive DoFs, while balancing costs and constraints for real-world feasibility and performance. However, their compatibility with the design of robot legs with multiple active DoFs remains unclear. Additionally, the simultaneous optimization is highly nonlinear and computationally intensive, which may make convergence to optimal designs difficult. Lastly, having an explicit list of available topologies not only enables filtering out undesired ones to reduce the search space and improve convergence, but also naturally provides multiple candidate designs for comparison and selection based on criteria difficult to encode during optimization.

III. METHODOLOGY

The proposed tool for synthesizing planar linkages with revolute joints consists of two major components: topology enumeration and dimensions optimization. First, multi-DoF topologies with analytical forward kinematics are enumerated with graph operations. Second, the dimensions for a particular topology are found through formulating and solving an optimization problem with costs and constraints that specify the design requirements.

A. Topology Enumeration

1) *Generation*: The generation of a topology is achieved by starting with a ground link and then repeatedly applying one of two operations to the previous topologies. The first operation M adds a link via a motorized revolute joint whose angle can be specified arbitrarily. This operation increases the linkage's DoFs by 1. The second operation F adds a pair of follower links via two additional passive revolute joints, which does not affect the linkage's DoFs. The sequence of operations or the plan $\mathcal{O} = \{O_0, O_1, \dots, O_N\}$, $O \in \{M, F\}$ defines a topology, and various topologies can be generated by changing the order and number of operations applied.

Fig. 2 (a1)-(a7) show some example topologies that are possible to generate in their structural representations, where the links are polygons, and the joints are dots [9]. Examples

include multi-DoF serial chains, one-DoF four-bar and six-bar linkages, and a two-DoF five-bar linkage. Fig. 2(b1)-(b7) show their corresponding graph representations, where the links are the nodes, and the joints are the edges [9].

In graph representations, each node has a unique index for specifying attachment locations of the followers. Each node also has a label: ground (g), link (l), or output (o). Each edge can either be labeled as the motor (m) or the joint (j). These labels help distinguish different child topologies of a parent topology. For example, depending on whether the coupler or the rocker of a four-bar linkage is labeled as the output, the output link can either only rotate around a point or perform more complicated trajectories. These labels are also essential for formulating the dimensional optimization, such as selecting topologies with a certain number of motorized joints, and placing constraints on ground and output joints. These graph representations are also essential for the enumeration process.

2) *Enumeration*: Following this topology generation method, all the possible topologies can be enumerated with a tree as shown in Fig. 3. The tree starts from an initial graph G_0 containing nodes 0 and 1 connected with an edge $[0, 1]$.

To grow the tree, there are two possible types of operations: adding a motorized link $G' = M(G, i, a)$ that stands for attaching a new node a to node i of graph G with a new edge $[i, a]$ to form a new graph G' ; and adding a pair of follower links $G' = F(G, (i, j), (a, b))$ that stands for attaching two new nodes a and b connected with an edge $[a, b]$ to node i and j of graph G through two new edges $[i, a]$ and $[j, b]$ to form a new graph G' . For operation M , the possible options are simply attaching to existing nodes. For operation F , the possible options are attaching to the permutations of selecting two nodes from the existing nodes. The tree can be grown to an arbitrary depth through repeatedly applying these two operations.

Labeling is performed to each node or graph in the enumeration tree to represent a possible linkage topology. The first node is labeled as the ground. The last node is labeled as the output. The other nodes are labeled as links. The edges are labeled as either joints or motors based on the operations applied.

To be noted, technically, the tree should start from a graph with a single ground node and branches out into performing operation M or F . However, adding follower links to the ground is redundant, and, therefore, the tree can safely start from a ground with a motorized link G_0 . Additionally, instead of the graph G , the plan $\mathcal{O}$ is used to represent a linkage topology, since the sequence is required for calculating forward kinematics and the graph can be constructed from it if needed.

3) *Isomorphism*: The topologies in the enumeration tree include both isomorphism and subgraph isomorphism. An example of isomorphism is given by the topologies in Fig. 4(a) and (b), where their only difference is the attachment order of node 4 and 5, which does not affect the output motion. An example of subgraph isomorphism is given by Fig. 4(c) and (d), where the topology in Fig. 4(c) and a

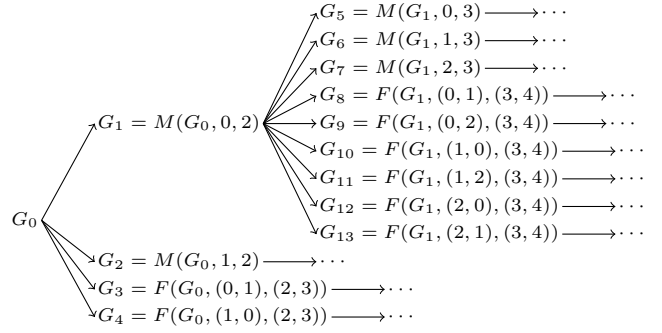
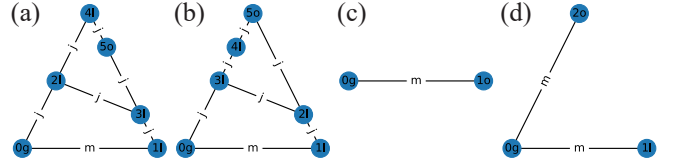


Fig. 3. Tree for enumerating all possible topologies. After labeling, G_0 is the graph in Fig. 2(b1); G_1 is Fig. 2(b4); G_2 is Fig. 2(b2); G_3 is Fig. 2(b3); and G_{11} is Fig. 2(b7).



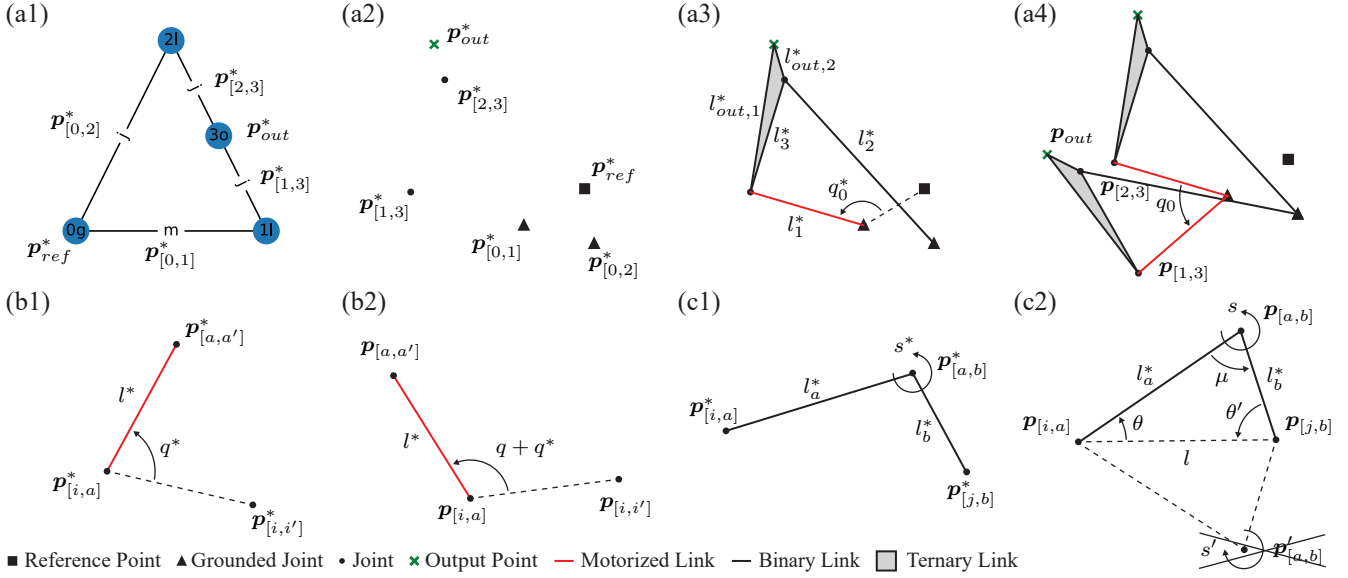


Fig. 5. Forward kinematics of linkage topologies. (a1)-(a4) Steps of calculating forward kinematics demonstrated with a four-bar linkage: (a1) determine the number of points needed to define the dimensions of a linkage design; (a2) sample or manually specify the points; (a3) calculate the dimensions from the points; and (a4) calculate new point locations given the angle of the motorized joints. (b1) and (b2) Calculating the dimensions and new joint location of operation M . (c1) and (c2) Calculating the dimensions and new joint location of operation F .

constant, include the grounded joints $\{p_{[i,j]}^* \mid i = 0 \vee j = 0\}$, the reference point p_{ref}^* , the link lengths l^* , the orders of the pairs of follower links s^* , and the home angles of the motorized joints q^* . Given the dimensions and the angles of the motorized joints q , the locations of all the non-grounded joints $\{p_{[i,j]} \mid i \neq 0 \wedge j \neq 0\}$ and the output point p_{out} can also be calculated. Note that the superscript $*$ denotes driving variables and driven variables do not have it.

For operation $G' = M(G, i, a)$, as illustrated in Fig. 5(b1), the dimensions include the link length of the motorized link l^* and the home angle of its joint q^* , which are

$$l^* = \|p_{[a,a']}^* - p_{[i,a]}^*\|, \quad (1)$$

$$q^* = \text{atan2}(v_y, v_x) - \text{atan2}(v'_y, v'_x), \quad (2)$$

$$v = p_{[a,a']}^* - p_{[i,a]}^*, \quad (3)$$

$$v' = p_{[i,a]}^* - p_{[i,i']}^*, \quad (4)$$

where $p_{[a,a']}^*$ is another point on the motorized link and $p_{[i,i']}^*$ is another point on the reference link, which are retrieved from the respective graph. When calculating forward kinematics as illustrated in Fig. 5(b2), the new location of the joint on the motorized link is

$$p_{[a,a']} = [l^* \cos(q_{total}), l^* \sin(q_{total})] + p_{[i,a]}, \quad (5)$$

$$q_{total} = q + q^* + \text{atan2}(v_y, v_x), \quad (6)$$

$$v = p_{[i,a]}^* - p_{[i,i']}^*, \quad (7)$$

where $p_{[i,a]}$ and $p_{[i,i']}$ are the new locations of points on the reference link, which are known because they either have already been calculated by previous operations, or they are on the ground link.

For operation $G' = F(G, (i, j), (a, b))$, as illustrated in Fig. 5(c1), the dimensions include the link lengths of the follower links l_a^* and l_b^* as well as their order s , which are

$$l_a^* = \|p_{[a,b]}^* - p_{[i,a]}^*\|, \quad (8)$$

$$l_b^* = \|p_{[a,b]}^* - p_{[j,b]}^*\|, \quad (9)$$

$$s^* = \text{sign}((p_{[a,b]}^* - p_{[i,a]}^*) \times (p_{[a,b]}^* - p_{[j,b]}^*)), \quad (10)$$

where the order of the two links is determined through the sign of the cross product between their respective vectors. When calculating forward kinematics as illustrated in Fig. 5(c2), the new location of the joint connecting the two follower links is

$$p_{[a,b]} = lv/\|v\| + p_{[i,a]}, \quad (11)$$

$$v = R(\theta)(p_{[j,b]} - p_{[i,a]}), \quad (12)$$

where $R(\theta)$ is the rotation matrix and the angle θ can be solved using the law of cosines,

$$\cos(\theta) = (l_a^{*2} + l_b^{*2} - l^2)/2l_a^*l_b^*, \quad (13)$$

$$l = \|p_{[i,a]} - p_{[j,b]}\|, \quad (14)$$

$$\theta = s^* \cdot \cos^{-1}(\cos(\theta)), \quad (15)$$

where s^* is used to select the solution that does not alter the order of the two links, and $p_{[i,a]}$ and $p_{[j,b]}$ are the new locations of points on the reference links, which are known for the same reason mentioned for operation M .

Equations (8)-(15) are reused to calculate the forward kinematics of the location of the output point p_{out} , because its relationship with the output link can be viewed as a special case of operation F , with both follower links connected to the same link resulting in a rigid connection and not adding new links. Additionally, this also applies to the forward kinematics of links with more than two joints, where the relationship between an additional joint and the existing ones is described in the same way.

In summary, the forward kinematics of a linkage topology G represented by its plan O is denoted as $p = FK(p^*, q)$, where p^* is a list of points defining the home configuration of the linkage, q is the joint angles of the motorized links, and

$\mathbf{p}$ contains the resulting joint, reference, and output locations. It is computed based on a sequence of calculations that are purely analytical $\mathcal{C} = \{C_0, C_1, \dots, C_N\}$, $C \in \{C_M, C_F\}$, where C_M represents equations (1)-(7) and C_F represents equations (8)-(15). For each calculation, the related point keys and dimensions are precomputed. The sequence of calculations $\mathcal{C}$ is derived from the plan $\mathcal{O}$, but has more steps to handle the links with more than two joints and the output link. To be noted, both $\mathbf{q}$ and $\mathbf{p}$ have an extra dimension to accommodate a design's multiple configurations for different combinations of motorized joint angles.

2) *Optimization Formulation*: Using the forward kinematics, the dimensional synthesis of a linkage topology is formulated as an optimization problem with the following general form:

$$\min_{\mathbf{p}^*, \mathbf{J}_{\text{ref}}, \mathbf{q}_r} \mathbf{w}^\top \left[f_1(\mathbf{p}, \mathbf{c}) \cdots f_{N_f}(\mathbf{p}, \mathbf{c}) \quad \sum \sigma_{\mathbf{q}_r, k}^2 \right]^\top \quad (16)$$

$$\text{subject to } |\cos([\boldsymbol{\theta} \quad \boldsymbol{\theta}' \quad \boldsymbol{\mu}])| < 1 - \epsilon_{\cos}, \quad (17)$$

$$\lambda_{\min}(\mathbf{J}\mathbf{J}^\top)/\lambda_{\max}(\mathbf{J}\mathbf{J}^\top) > \epsilon_{\mathbf{J}\mathbf{J}^\top}, \quad (18)$$

$$g_j(\mathbf{p}, \mathbf{c}) < 0, j = 1, \dots, N_g \quad (19)$$

$$\text{where } (\mathbf{p}, \cos([\boldsymbol{\theta} \quad \boldsymbol{\theta}' \quad \boldsymbol{\mu}])) = FK(\mathbf{p}^*, \mathbf{q}), \quad (20)$$

$$\mathbf{q} = \mathbf{J}_{\text{ref}}^{-1} \mathbf{x}^d + \mathbf{q}_r, \quad (21)$$

$$\dot{\mathbf{x}} = \mathbf{J} \dot{\mathbf{q}}. \quad (22)$$

This optimization mainly tries to find a linkage's home configuration $\mathbf{p}^*$ that minimizes the weighted sum of cost functions $\sum \mathbf{w}_i f_i$ under constraint functions g_i that specify the design requirements. Both the cost and constraint functions depend on the linkage's configurations $\mathbf{p}$ and some constants $\mathbf{c}$ such as the desired output locations and constraint boundaries.

One of the challenges for synthesizing a multi-DoF design is that, for a set of desired outputs $\mathbf{x}^d$, the input joint angles $\mathbf{q}$ are difficult to specify, because the exact mapping between them are often unknown and the forward kinematics in (20) can not be calculated without $\mathbf{q}$. To overcome this, two more groups of optimizable variables are introduced: a reference Jacobian $\mathbf{J}_{\text{ref}}$ constant across outputs, and residuals $\mathbf{q}_r$ for all the input angles. As in (21), they essentially enable optimizing for $\mathbf{q}$ indirectly. The reason behind not directly treating $\mathbf{q}$ as optimizable variables is that, without $\mathbf{J}_{\text{ref}}$, the optimized results will almost certainly have inputs on different kinematic branches [23], similar to the inputs jumping from "elbow up" to "elbow down" poses for serial arms. Formulating $\mathbf{q}$ this way keeps the mapping close to a constant-Jacobian design, but still allows some nonlinearity via $\mathbf{q}_r$. The degree of nonlinearity allowed is tuned via a cost that penalizes the variance of the residuals for each individual input joint $\sigma_{\mathbf{q}_r, k}^2$.

Two constraints can be used for avoiding singularities. The angles $[\boldsymbol{\theta} \quad \boldsymbol{\theta}' \quad \boldsymbol{\mu}]$ in Fig. 5(c2) should not be close to 0 or π rad. For multi-DoF designs, a constraint on the condition number of the matrix $\mathbf{J}\mathbf{J}^\top$ for the manipulability ellipsoid [24] is necessary too. The inverse of the condition number is used to avoid large values. Notably, since the entries of $\mathbf{J}$

may have different units, they should be scaled to similar ranges with constant factors before calculation. Since the forward kinematics is analytical, the actual Jacobian $\mathbf{J}$ of the design can be derived without approximation.

C. Implementation

This entire linkage synthesis pipeline, including the topology enumeration and dimensional optimization, is implemented in Python. NetworkX is used to construct and manipulate graphs. It has built-in functions for checking isomorphism and subgraph isomorphism. Some parallelization on a central processing unit (CPU) is implemented to speed up the isomorphism removal process.

PyTorch is used for calculating forward kinematics and optimizing dimensions, as it automatically keeps track of gradients for optimization. The optimization constraints are treated as highly weighted costs that are clamped to zero once the requirements are met, so that they can be handled the same way as regular costs. Moreover, not all $\mathbf{q}$ is valid for forward kinematics, which is indicated by $\cos(\boldsymbol{\theta})$ in (13) exceeding 1. To prevent interruption to calculating forward kinematics, its value clipped to $(-1, 1)$ is used for downstream calculations, and the constraint for the original value (17) is relied upon for kinematically valid designs.

To escape local minima during optimization, multiple home configurations $\{\mathbf{p}_0^*, \mathbf{p}_1^*, \dots\}$ are sampled and optimized simultaneously to select the one with the minimum cost. They are treated as a batch dimension and accelerated by parallel computation on a GPU. Since the dimensions optimization is formulated in a topology-independent way, utilizing the labels of graph nodes and links, the same code is reused to sweep over all candidate topologies to find the best design.

IV. EXAMPLES

In this section, three design examples are presented to showcase the capability of the proposed tool. The topology enumeration tree is expanded to the 6th layer. Only topologies with at most three F operations are candidates for further filtering, since too many follower links compromise real-world feasibility. For all examples, the points for the home configurations $\mathbf{p}^*$ were uniformly sampled from a 300 mm by 300 mm box. If present, entries of $\mathbf{J}_{\text{ref}}$ were sampled from either $(-1, 1)$ rad or $(-100, 100)$ mm¹, and $\mathbf{q}_r$ was sampled from $(-0.1, 0.1)$ rad. The desired workspace for each design is discretized into 25 points, a relatively low resolution to prioritize computation speed. Yet, no discontinuity was encountered for the prototypes of the optimized designs.

A. One-DoF Leg

As shown in Fig. 6(a), the first design is a one-DoF leg with a foot trajectory suitable for walking mechanisms. The step length is about 100 mm and the lift is about 25 mm. One-DoF topologies, with up to 6 links and a serial chain of at least 2 links after the ground to the output, are considered,

¹If $\mathbf{x}^d$ is in polar coordinates, its entries will have different units.

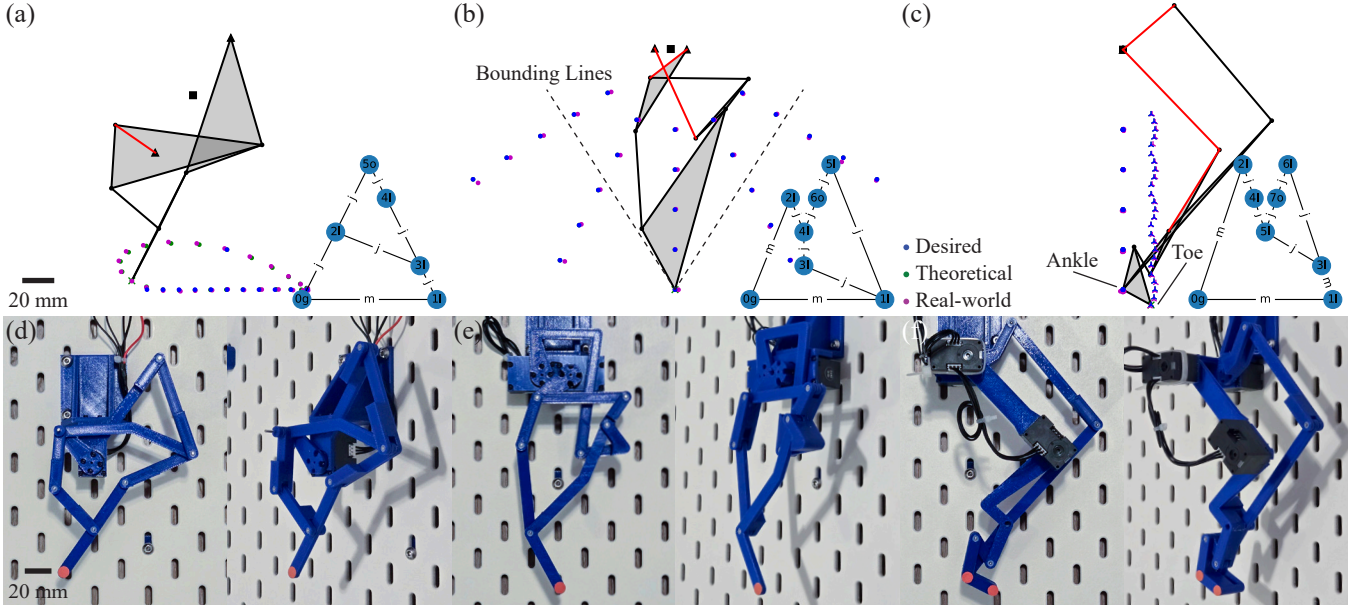


Fig. 6. Example leg designs and their topologies and prototypes. (a) and (d) One-DoF leg for walking mechanisms. (b) and (e) Two-DoF parallel leg with a fan-shaped workspace. (c) and (f) Three-DoF hybrid leg with an active ankle. The real-world results from the validation experiment are plotted alongside the desired and theoretical output locations.

because six-bar linkages provide a large design space and a single link can only generate circular trajectories. The dimensions optimization is formulated as

$$\min_{\mathbf{p}^*} \mathbf{w}^\top [e_{\text{out}} \quad l_{\text{total}}]^\top \quad (23)$$

$$\text{subject to } |\cos([\theta \quad \theta' \quad \mu])| < 0.8, \quad (24)$$

$$\|\mathbf{p}_{\text{joint},i,n_{\text{NG}}} - \mathbf{p}_{\text{joint},j,n_{\text{NG}}}\| > 20 \text{ mm}, \quad (25)$$

$$\mathbf{y}_{\text{other}} - \mathbf{y}_{\text{out}} > 20 \text{ mm}, \quad (26)$$

$$|\mathbf{x}_{\text{joint}}| < 75 \text{ mm}, \quad (27)$$

$$\text{where } e_{\text{out}} = 1/N_{\mathbf{p}_{\text{out}}^d} \sum \min(\|\mathbf{p}_{\text{out}} - \mathbf{p}_{\text{out},i}^d\|). \quad (28)$$

The main cost is the output trajectory error e_{out} . As shown in Fig. 6(a), the desired output trajectory $\mathbf{p}_{\text{out}}^d$ uses 11 points to specify a straight and constant-speed foot movement and 1 point to require a foot lift. 25 output points $\mathbf{p}_{\text{out}}$ are calculated with forward kinematics given an evenly-spaced full input rotation $\mathbf{q}$. e_{out} specified as (28) essentially makes sure that each desired output point has at least one matching point from the actual output points, where $\mathbf{p}_{\text{out},i}^d$ is a matrix tiled with the i -th row of $\mathbf{p}_{\text{out}}^d$ to the same size of $\mathbf{p}_{\text{out}}$. In this way, the exact lifting trajectory is not required, allowing for more potential designs. This also removes the need of optimizing $\mathbf{J}_{\text{ref}}$ and $\mathbf{q}_r$. The secondary cost is the total link length of the linkage l_{total} to keep the design compact. The length of each link is defined as the distance from each joint to the link's centroid, which applies to links with more than two joints and is easy to compute.

Several constraints are also necessary: 1) a conservative threshold of 0.8 for (24) is imposed, because the intended fabrication method does not use parts with tight tolerances and very stiff materials; 2) the distance between joints on a non-grounded link $\mathbf{p}_{\text{joint},i,n_{\text{NG}}}$ should not be too small to make space for the bushing and avoid multiple layers on a

single link as in (25); 3) some vertical clearance between the output and the other points is enforced in (26); and 4) the linkage's width is constrained around the origin as in (27).

B. Two-DoF Parallel Leg

As shown in Fig. 6(b), the second design is a two-DoF parallel leg specified by a fan-shaped workspace, which is inspired by a swinging pogo stick, commonly used for legged locomotion modeling. The swinging angle is 2 rad and the stroke length is 100 mm. Two-DoF topologies, with up to 7 links, a serial chain of at least 2 links after the ground to the output, and only 2 grounded joints that are also motorized, are considered. The exclusion of more grounded joints aims to reduce the complexity of the physical prototypes. The dimensions optimization is formulated as

$$\min_{\mathbf{p}^*, \mathbf{J}_{\text{ref}}, \mathbf{q}_r} \mathbf{w}^\top [e_{\text{out}} \quad l_{\text{total}} \quad \sum \sigma_{\mathbf{q}_{r,k}}^2]^\top \quad (29)$$

$$\text{subject to } |\cos([\theta \quad \theta' \quad \mu])| < 0.8, \quad (30)$$

$$\lambda_{\min}(\mathbf{J}\mathbf{J}^\top)/\lambda_{\max}(\mathbf{J}\mathbf{J}^\top) > 0.05, \quad (31)$$

$$\|\mathbf{p}_{\text{joint},i,n} - \mathbf{p}_{\text{joint},j,n}\| > 20 \text{ mm}, \quad (32)$$

$$(\mathbf{y}_{\text{other}}^{\mathbf{R}} - \mathbf{y}_{\text{ankle}}^{\mathbf{R}})/|\mathbf{x}_{\text{other}}^{\mathbf{R}}| > 1 \text{ rad}, \quad (33)$$

$$\max(\mathbf{x}_{\text{joint}}^{\mathbf{R}}) - \min(\mathbf{x}_{\text{joint}}^{\mathbf{R}}) < 150 \text{ mm}, \quad (34)$$

$$\text{where } e_{\text{out}} = 1/N_{\mathbf{p}_{\text{out}}^d} \sum \|\mathbf{p}_{\text{out}} - \mathbf{p}_{\text{out}}^d\|. \quad (35)$$

The main cost in this example is also the output trajectory error e_{out} , but one-to-one matches are preferred as in (35). The shaping of the linkage requires some modifications compared to the previous example: 1) the grounded joints should have some minimum clearance between each other so that the motors can be placed side by side as in (32) 2) the points other than the output should stay within an inverted isosceles triangle pointing at the output, as expressed in (33) and illustrated with the bounding lines in Fig. 6(b); and 3) the

width of the entire linkage is also constrained, but does not need to be centered around the origin as in (34). These latter two constraints are applied after the linkage is rotated around the origin to the upright form at each output configuration, denoted by the superscript $\mathbf{R}$.

C. Three-DoF Hybrid Leg

As shown in Fig. 6(c), the third design is a three-DoF series-parallel hybrid leg with an active ankle specified by both the desired ankle and toe locations. The workspace of the ankle is a 100 mm straight line for leg extension and retraction. The toe can then rotate 1 rad about the ankle at each point. To simplify the problem, it is assumed that all the grounded joints are at the origin, which makes the swinging of the leg trivial. Three-DoF topologies, with up to 8 links, a serial chain of at least 3 links after the ground to the output, only 2 grounded and motorized joints, and 2 connected motorized links, are considered. The last requirement excludes topologies in which the motor is directly at the ankle joint. For this design, the toe is the output point, but the ankle joint can be one of the other joints on the output link. Thus, more child topologies exist for the dimensions to be optimized over. The dimensions optimization is formulated as

$$\min_{\mathbf{p}_{\text{NG}}^*, \mathbf{J}_{\text{ref}}, \mathbf{q}_r} \mathbf{w}^\top \left[e_{\text{ankle}} \quad e_{\text{toe}} \quad l_{\text{total}} \quad \sum \sigma_{\mathbf{q}_{r,k}}^2 \right]^\top \quad (36)$$

$$\text{subject to } |\cos([\boldsymbol{\theta} \quad \boldsymbol{\theta}' \quad \boldsymbol{\mu}])| < 0.8, \quad (37)$$

$$\lambda_{\min}(\mathbf{J}\mathbf{J}^\top)/\lambda_{\max}(\mathbf{J}\mathbf{J}^\top) > 0.05, \quad (38)$$

$$\|\mathbf{p}_{\text{joint},i,n_{\text{NG}}} - \mathbf{p}_{\text{joint},j,n_{\text{NG}}}\| > 20 \text{ mm}, \quad (39)$$

$$(\mathbf{y}_{\text{other}} - \mathbf{y}_{\text{ankle}})/|\mathbf{x}_{\text{other}}| > 0.5 \text{ rad}, \quad (40)$$

$$\max(\mathbf{x}_{\text{joint}}) - \min(\mathbf{x}_{\text{joint}}) < 150 \text{ mm}, \quad (41)$$

$$\text{where } e_{\text{ankle}} = 1/N_{\mathbf{p}_{\text{ankle}}^d} \sum \|\mathbf{p}_{\text{ankle}} - \mathbf{p}_{\text{ankle}}^d\|, \quad (42)$$

$$e_{\text{toe}} = 1/N_{\mathbf{p}_{\text{toe}}^d} \sum \|\mathbf{p}_{\text{toe}} - \mathbf{p}_{\text{toe}}^d\|. \quad (43)$$

Since grounded joints are at the origin, only the non-grounded joints of the home configuration $\mathbf{p}_{\text{NG}}^*$ are optimizable. The costs and constraints are similar to those in the previous examples. The main difference is the use of both ankle error e_{ankle} and toe error e_{toe} terms to specify the desired foot poses. Toe points are also not considered in the foot clearance and leg width.

D. Computation

The optimization was run on a workstation computer². 1,000 home configurations were sampled and optimized simultaneously on the GPU. The Adam optimizer was used in conjunction with a cosine learning rate scheduler for 10,000 steps with an initial learning rate of 1. To be noted, $\mathbf{J}_{\text{ref}}$ and $\mathbf{q}_r$ were scaled with constant factors so that their initial values are closer to that of $\mathbf{p}^*$, and the same learning rate can be used. Information related to the synthesis is listed in Table I. The best designs are shown in Fig. 6(a)-(c). The supplementary video provides a visualization of the dimensions optimization process, along with the top three designs for each example.

TABLE I
SYNTHESIS INFORMATION

Leg Design	One-DoF	Two-DoF Parallel	Three-DoF Hybrid
Max. # of Links	6	7	8
# of Topologies	9	6	34
Max. # of Opt. Var.	18,000	74,000	100,000
Seconds per Topology	16.1	49.4	66.0

E. Validation

As shown in Fig. 6(d)-(f), the prototypes of the best designs were fabricated for validation. They were designed using computer-aided design (CAD) software. The links were manually layered and their shapes adjusted to avoid collisions. Each joint was supported by two nylon bushings and a stainless steel pin press-fit into the link. Then, the links were 3D-printed with polylactic acid (PLA) and assembled with servos³. The legs were mounted on a pegboard, the servos were commanded to the input angles, and videos were taken. The actual locations of the output points were extracted from colored markers on both the output link and the board using computer vision. The results are plotted and compared against the desired ones in Fig. 6(a)-(c). Prototype demonstrations are included in the supplementary video too.

V. DISCUSSIONS

The successful synthesis of the three leg designs with different DoFs showcases the versatility of the proposed planar linkage synthesis tool. Various types of design requirements can be specified as costs and constraints. Moreover, the same optimization can be performed across different topologies without modifications, providing a large design space to explore. The gradient-based optimization is fast, efficient, and scalable, providing multiple design candidates within minutes. Lastly, the relatively good alignment between the physical prototypes' behaviors and the theoretical designs demonstrates that the designs generated by the proposed method are also viable in the real world.

It is worth noting that determining the exact optimization formulations that yield feasible and satisfactory designs is an iterative process that requires tuning. For example, the constraints for avoiding singularities (17) and (18) were added after discovering that some initial prototypes were locking internally or losing rigidity. The cost and constraint, along with their respective weights and bounds, also require some trial and error based on feedback from the optimization results. Fortunately, the proposed tool's speed and versatility provide ample design responsiveness and freedom to the process.

The proposed tool also has some limitations. Besides the exclusion of prismatic joints, not all revolute-joint-only topologies can be generated, such as the Stephenson II one-DoF six-bar linkage [13] and 3-RRR three-DoF eight-bar parallel linkage [9]. However, we are confident that additional topologies can be included by revising the labeling

²Intel Core i9-13900K, NVIDIA GeForce RTX 4090, Ubuntu 22.04

³ROBOTIS XL330-M288-T, XC330-T181-T

method and the optimization formulation, since inversions of some topologies, such as those mentioned above, have analytical forward kinematics and are already covered by the current tool. Another limitation is that only topologies with a single output link are considered, which may be insufficient when the design requirements of the intermediate links are also present. There should be ways around it, just like how each topology is further divided into child ones for the three-DoF hybrid leg example. Nevertheless, a more formal way to handle multiple outputs is still worth exploring. Finally, even though the “motorized” joints are treated as actuated joints in all examples, they can, in theory, also represent passive joints or joints with springs. More designs have to be attempted to verify this function.

VI. CONCLUSIONS

In this paper, a topological and dimensional synthesis tool is proposed for multi-DoF planar linkages with revolute joints. By combining enumeration of topologies with analytical forward kinematics and gradient-based optimization for finding suitable dimensions, it provides a versatile, flexible, fast, and scalable approach to designing and tuning robot legs for specific applications. Three examples are presented to showcase these advantages and to provide some guidelines and inspiration for how designs can be handled.

Future work will focus on improving the tool’s capability. Support for prismatic joints will be added by expanding the operation set to include adding motorized prismatic joints and follower pairs containing passive prismatic joints. Support for spherical linkages should also be feasible, since their topologies are identical to those of planar ones. The major difference will be how forward kinematics is calculated. Costs and constraints that provide more advanced linkage shaping for automatic CAD generation and achieving complex functions will be evaluated and added. We also believe that the concept of enumerating topologies with analytical forward kinematics and using gradient-based optimization to determine dimensions is applicable to spatial linkage synthesis. The main challenge will be defining the set of operations for generating spatial topologies.

Despite some limitations, the proposed tool has significantly broadened the accessible design space of robot legs. It will be used to develop novel legged robots tailored for different applications. This tool’s potential for grippers, wings, arms, and more will also be explored.

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